Chapter 21 Electromagnetic Waves

21.1 The equation of wave motion

Maxwell's equations are

$$\nabla \cdot \boldsymbol{D} = \rho_{\text{free}},$$

$$\nabla \cdot \boldsymbol{B} = 0,$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t},$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{j}_{\text{free}} + \frac{\partial \boldsymbol{D}}{\partial t}.$$

If there are no free charge and current, i.e.

$\boldsymbol{j}_{\text{free}} = 0, \rho_{\text{free}} = 0,$ Maxwell's equations becomes

$$\nabla \cdot \boldsymbol{D} = 0,$$
$$\nabla \cdot \boldsymbol{B} = 0,$$
$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t},$$
$$\nabla \times \boldsymbol{H} = \frac{\partial \boldsymbol{D}}{\partial t}.$$

In the vacuum

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E},$$
$$\boldsymbol{B} = \mu_0 \boldsymbol{H}.$$

$$\Longrightarrow$$

$$\nabla \cdot \boldsymbol{E} = 0,$$
$$\nabla \cdot \boldsymbol{B} = 0,$$
$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t},$$
$$\nabla \times \boldsymbol{B} = \varepsilon_0 \mu_0 \frac{\partial \boldsymbol{E}}{\partial t}.$$

Taking rotation on both sides of the 3rd equation, we have

$$\nabla \times (\nabla \times \boldsymbol{E}) = -\nabla \times \frac{\partial \boldsymbol{B}}{\partial t}.$$

left side: $\nabla\times(\nabla\times\boldsymbol{E})=\nabla(\nabla\cdot\boldsymbol{E})-\nabla^2\boldsymbol{E}=-\nabla^2\boldsymbol{E}$

right side: $-\nabla\times\frac{\partial\boldsymbol{B}}{\partial t}=-\frac{\partial}{\partial t}(\nabla\times\boldsymbol{B})=-\varepsilon_0\mu_0\frac{\partial^2\boldsymbol{E}}{\partial t^2}$

$$\Downarrow$$

Wave equation for the electric field: $\nabla^2\boldsymbol{E}-\varepsilon_0\mu_0\frac{\partial^2\boldsymbol{E}}{\partial t^2}=0.$

We can obtain from a similar derivation that

$$\nabla^2\boldsymbol{B}-\varepsilon_0\mu_0\frac{\partial^2\boldsymbol{B}}{\partial t^2}=0.$$

The velocity of the propagating wave is

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}.$$

the velocity of light
in the vacuum

$$\nabla^2 \boldsymbol{E} - \varepsilon_0 \mu_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} = 0.$$

The wave equation of electromagnetic field in the nonconducting medium

In the medium, if the electric and magnetic fields oscillate with a definite frequency ω, we have

$$\boldsymbol{D} = \varepsilon(\omega)\boldsymbol{E},$$

$$\boldsymbol{B} = \mu(\omega)\boldsymbol{H}.$$

Then, we express the time dependent electromagnetic field as

$$\boldsymbol{E}(\boldsymbol{r},t) = \boldsymbol{E}(\boldsymbol{r})e^{-i\omega t},$$

$$\boldsymbol{B}(\boldsymbol{r},t) = \boldsymbol{B}(\boldsymbol{r})e^{-i\omega t}.$$

The actual quantities are Re$\boldsymbol{E}$ and Re$\boldsymbol{B}$.

the monochrome wave

Substituting above expressions into Maxwell's equations, we have

$$\nabla \cdot \boldsymbol{D} = 0, \qquad \nabla \cdot \boldsymbol{E}(\boldsymbol{r}) = 0,$$

$$\nabla \cdot \boldsymbol{B} = 0, \qquad \nabla \cdot \boldsymbol{B}(\boldsymbol{r}) = 0,$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \qquad \Longrightarrow \qquad \nabla \times \boldsymbol{E}(\boldsymbol{r}) = i\omega \boldsymbol{B}(\boldsymbol{r}),$$

$$\nabla \times \boldsymbol{H} = \frac{\partial \boldsymbol{D}}{\partial t}. \qquad \nabla \times \boldsymbol{B}(\boldsymbol{r}) = -i\omega\varepsilon(\omega)\mu(\omega)\boldsymbol{E}(\boldsymbol{r}).$$

Taking rotation on both sides of the 3rd equation, we obtain

$$\nabla \times (\nabla \times \boldsymbol{E}(\boldsymbol{r})) = i\omega \nabla \times \boldsymbol{B}(\boldsymbol{r}) = \omega^2 \varepsilon\mu \boldsymbol{E}(\boldsymbol{r}).$$

or

$$\nabla^2 \boldsymbol{E}(\boldsymbol{r}) + k^2 \boldsymbol{E}(\boldsymbol{r}) = 0, \qquad k^2 = \omega^2 \varepsilon\mu.$$

Helmholtz equation

Similarly, we can obtain

$$\nabla^2 \boldsymbol{B}(\boldsymbol{r}) + k^2 \boldsymbol{B}(\boldsymbol{r}) = 0.$$

Helmholtz equation---the differential equation of wave motion at definite frequency.

21.2 Propagation of the electromagnetic wave

plane wave
spherical wave
cylindrical wave

wavefront or wave surface

Assume the electromagnetic wave is propagating along the x axis. Then, ***E*** and ***B*** are only functions of *x* and *t*. Thus, the Helmholtz equation becomes

$$\frac{d^2}{dx^2}\boldsymbol{E}(x) + k^2\boldsymbol{E}(x) = 0,$$

And one of its solutions is

$$\boldsymbol{E}(x) = \boldsymbol{E}_0 e^{ikx}.$$

Therefore

$$\boldsymbol{E}(x,t) = \boldsymbol{E}_0 e^{i(kx-\omega t)}.$$

The actual electric field is

$$\boldsymbol{E}(x,t) = \boldsymbol{E}_0 \cos(kx-\omega t) = \boldsymbol{E}_0 \cos\left[k\left(x-\frac{\omega}{k}t\right)\right].$$

The wavefront(or the wave surface--the surface on which all the point have the same phase) is x=Const.

The velocity of the wave (the phase velocity) is

$$v = \frac{\omega}{k} = \frac{1}{\sqrt{\varepsilon\mu}}.$$

$$\varphi = kx - \omega t, \quad \frac{d\varphi}{dt} = 0 = k\frac{dx}{dt} - \omega. \quad \Longrightarrow \quad v_p = \frac{dx}{dt} = \frac{\omega}{k}.$$

$$v = \frac{\omega}{k} = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{1}{\sqrt{\varepsilon_0\mu_0\varepsilon_r(\omega)\mu_r(\omega)}}$$

$$= \frac{c}{\sqrt{\varepsilon_r(\omega)\mu_r(\omega)}}.$$

The index of refraction of the medium is

$$n = \frac{c}{v} = \sqrt{\varepsilon_r(\omega)\mu_r(\omega)}.$$

dispersion

Since $\mu_r(\omega) \sim 1$ for majority of substances, then

$$n = \sqrt{\varepsilon_r(\omega)}.$$

The plane wave propagating along an arbitrary direction can be written as

$$\boldsymbol{E}(\boldsymbol{r},t) = \boldsymbol{E}_0 e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)},$$

where $\boldsymbol{k}$ is called the wave vector.

The wavefronts $\boldsymbol{k}\cdot\boldsymbol{r} = \text{Const.}$

$k = |\boldsymbol{k}|$ is called wave number.

$$k = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi}{vT} = \frac{2\pi}{\lambda}.$$

Substituting $\boldsymbol{E}(x,t) = \boldsymbol{E}_0 e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}$ into $\nabla\cdot\boldsymbol{E} = 0,$
we have

$$\nabla\cdot\boldsymbol{E} = \boldsymbol{E}_0\cdot\nabla e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}$$

$$= i\boldsymbol{k}\cdot\boldsymbol{E}_0 e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} = i\boldsymbol{k}\cdot\boldsymbol{E} = 0,$$

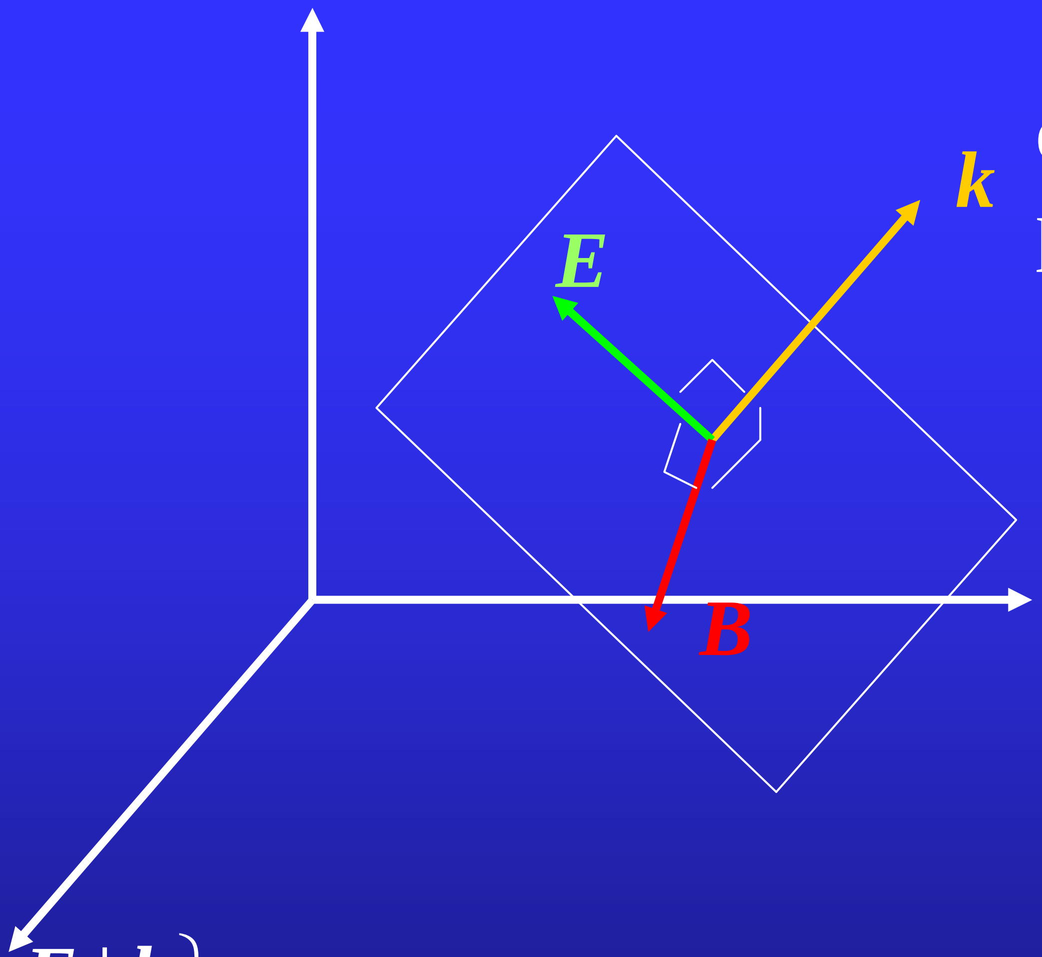
or $\boldsymbol{k}\cdot\boldsymbol{E} = 0.$

Similarly, we have $\boldsymbol{k}\cdot\boldsymbol{B} = 0.$

From $\nabla\times\boldsymbol{E}(\boldsymbol{r}) = i\omega\boldsymbol{B}(\boldsymbol{r}),$ we have

$$\boldsymbol{B} = \frac{1}{i\omega}\nabla\times\boldsymbol{E} = \frac{1}{i\omega}(i\boldsymbol{k}\times\boldsymbol{E})$$

$$= \frac{\boldsymbol{k}}{\omega}\times\boldsymbol{E} = \frac{1}{v}\hat{\boldsymbol{k}}\times\boldsymbol{E}.$$



$$\left.\begin{array}{l} \boldsymbol{E} \perp \boldsymbol{k} \\ \boldsymbol{B} \perp \boldsymbol{k} \end{array}\right\} \text{transverse wave}$$

$$\boldsymbol{E} \perp \boldsymbol{B}, \frac{|\boldsymbol{E}|}{|\boldsymbol{B}|} = v$$

The Poynting vector is

$$\boldsymbol{S} = \frac{1}{\mu}\boldsymbol{E}\times\boldsymbol{B} = \frac{1}{\mu v}\boldsymbol{E}\times(\hat{\boldsymbol{k}}\times\boldsymbol{E}) = \frac{1}{\mu v}E^2\hat{\boldsymbol{k}}$$

$$= \frac{E^2}{\mu v^2}v\hat{\boldsymbol{k}} = \varepsilon E^2 v\hat{\boldsymbol{k}} = \frac{1}{2}(\varepsilon E^2 + \frac{B^2}{\mu})v\hat{\boldsymbol{k}}.$$

In the vacuum,

$$\boldsymbol{S} = \frac{1}{2}(\varepsilon_0 E^2 + \frac{B^2}{\mu_0})c\hat{\boldsymbol{k}}.$$

The energy density is

$$u = \frac{1}{2}(\varepsilon_0 E^2 + \frac{B^2}{\mu_0}).$$

The momentum density is

$$\boldsymbol{g} = \frac{1}{c^2}\boldsymbol{S} = \frac{1}{2c}(\varepsilon_0 E^2 + \frac{B^2}{\mu_0})\hat{\boldsymbol{k}}.$$

The energy-momentum relation

$$u = gc\,.$$

Polarized wave

$$\boldsymbol{E} = E_0 \boldsymbol{e}_x e^{i(kz-\omega t)}.$$

linearly polarized wave

$$\boldsymbol{E} = E_0 (\boldsymbol{e}_x \pm i\boldsymbol{e}_y) e^{i(kz-\omega t)}.$$

circularly polarized wave

$$\mathrm{Re}\,\boldsymbol{E} = E_0 \boldsymbol{e}_x \cos(kz-\omega t) \mp E_0 \boldsymbol{e}_y \sin(kz-\omega t).$$

left-handed

right-handed

$$\boldsymbol{E} = (E_0 \boldsymbol{e}_x \pm iE_0' \boldsymbol{e}_y) e^{i(kz-\omega t)}, E_0 \neq E_0'.$$

elliptically polarized wave

Etinne Louis Malus   1809   reflection of setting sun from the windows of Luxembourg Palace   calcite

Polarization studies:

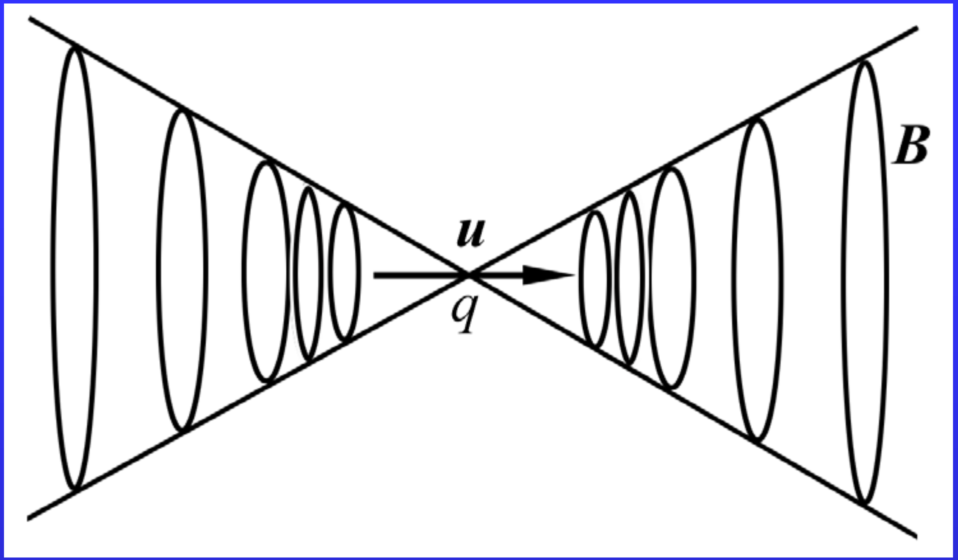
size and shape of virus particles

structure ranging in size from a galaxy (~10^{20}m) to a nucleus (~10^{-14}m)

polarizer

Brewster law

21.3 Radiation



$$E \sim \frac{q}{r^2}; \quad B \sim \frac{I}{\rho}$$

$$E' \sim \frac{q}{r^2} \qquad E \sim B \sim \frac{1}{r^2}; \quad S \sim \frac{1}{r^4}$$

$$\int_{r\to\infty} \boldsymbol{S} \cdot d\boldsymbol{A} \to 0$$

For an accelerated charge,

$$E \sim B \sim \frac{1}{r}; \quad S \sim \frac{1}{r^2}; \quad \oint S dA \neq 0.$$

For an oscillating electric dipole $\boldsymbol{p} = p_0 \boldsymbol{e}_z e^{-i\omega t}$,
when r>>λ, we have

$$\boldsymbol{E} = \frac{p_0 \sin\theta}{4\pi\varepsilon_0 r}\left(\frac{\omega}{c}\right)^2 e^{i(kr-\omega t)}\boldsymbol{e}_\varphi,$$

$$\boldsymbol{B} = \frac{p_0 \sin\theta}{4\pi\varepsilon_0 c r}\left(\frac{\omega}{c}\right)^2 e^{i(kr-\omega t)}\boldsymbol{e}_\theta.$$

The radiation energy current density is

$$\boldsymbol{S} = \frac{1}{\mu_0}\boldsymbol{E}\times\boldsymbol{B}$$

$$= \frac{p_0^{\,2}\sin^2\theta}{(4\pi\varepsilon_0 r)^2\mu_0 c}\left(\frac{\omega}{c}\right)^4 \cos^2(kr-\omega t)\boldsymbol{e}_r.$$

The average radiation energy current density is

$$S_{\text{ave}} = \frac{1}{T}\int_0^T S dt$$

$$= \frac{p_0^{\,2}\sin^2\theta}{2(4\pi\varepsilon_0 r)^2\mu_0 c}\left(\frac{\omega}{c}\right)^4.$$

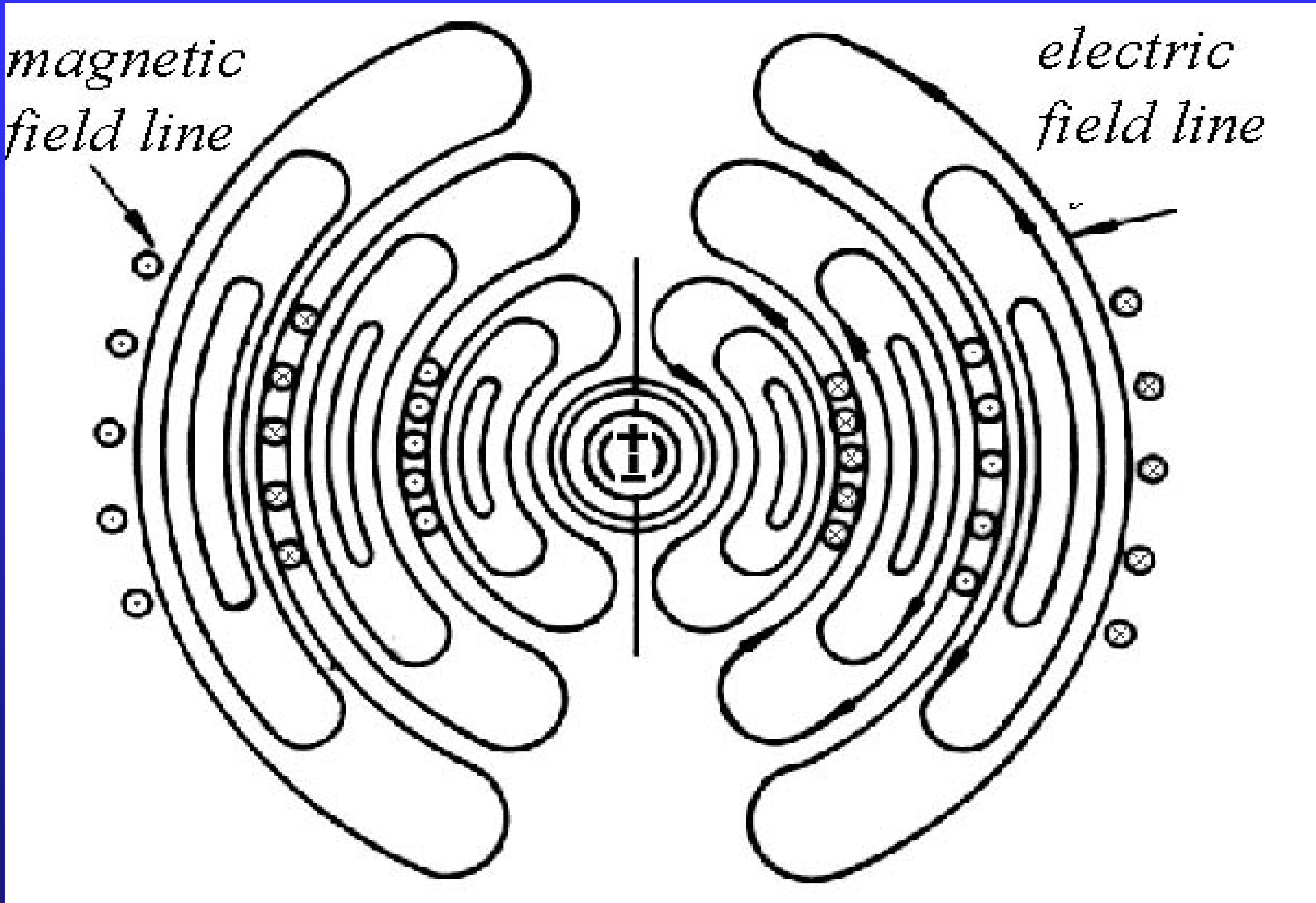
The total average energy radiated per unit time (the average power radiated) is

$$P = \oint S_{\mathrm{ave}} r^2 d\Omega$$

$$= \frac{p_0{}^2 \omega^4}{32\pi^2 \varepsilon_0 c^3} \oint \sin^2 \theta d\Omega$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{p_0{}^2 \omega^4}{3c^3}.$$

$$I = I_0 \cos \omega t$$



$$P \propto \omega^4.$$

Rayleigh scattering

Leonardo da Vinci
Mona Lisa
The Last Supper

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Radiations from space

The origin of radio astronomy In 1931, Karl Jansky, an American engineer at the Bell Telephone Laboratories (now Bell Laboratories, part of Lucent Technologies). in New Jersey, studied static that was interfering with short-wave communication systems. He found that the static appeared four minutes earlier each day. Jansky knew that the stars rise four minutes earlier daily, and so he concluded that the static must be coming from beyond the solar system. He was actually receiving radio waves from the center of our galaxy.

Professional astronomers did not follow up on Jansky's discovery. However, an American amateur astronomer named Grote Reber designed a radio telescope and operated it in his backyard beginning in the late 1930's. Radio astronomy finally began to flourish after World War II (1939-1945). The study of radio waves from space greatly expanded astronomers' knowledge of the structure, size, and history of the universe. For example, it revealed a great deal of new information about the clouds of gas and dust between the stars in our galaxy. During the 1960's, radio astronomers played an important role in the discovery of quasars and pulsars. In 1965, astronomers testing a radio telescope and receiver system discovered the primordial background radiation, which astronomers believe was produced when the universe began with the big bang.

Cosmic microwave background radiation In 1964, Arno Penzias(German-born American Physicist) and Robert Woodrow Wilson (American Physicist) were working at Bell Telephone Laboratories. They were using a type of radio telescope known as horn antenna, and trying to improve the radio-transmitted telephone signals by eliminating all the known sources of radio waves. But they were still left with a residue that was surprisingly isotropic. They tried to measure the signal at different times of day and on different days, but the signal did not vary.
Remnants of the electromagnetic radiation from the primeval fire ball
1979 Nobel prize

In 1967, A graduate student at the University of Cambridge Susan Jocelyn Bell and her adviser Antony Hewrish first observed a pulsar. While using a new radio telescope, she noted pulses coming from an unknown source, later identified as a pulsar. Her observations, and further work with Cambridge astronomer Antony Hewish, proved the existence of neutron stars. Since the 1930's, astronomers had believed that neutron stars existed. The discovery of pulsars gave rise to many new ideas about the origin and development of stars.

Antony Hewrish 1974 Nobel laureate

pulsars   rapidly spinning neutron stars   dense remains of burnt-out supergiant stars.

Scientists believe pulsars are rapidly spinning neutron stars, dense stars composed primarily of tightly packed neutrons, or perhaps of elementary particles called quarks. An extremely powerful magnetic field surrounds the neutron star and rotates with it. This rapidly rotating magnetic field produces a strong electric field that rips electrons and protons from the star's surface. As these particles flow from the star, they emit energy in the form of a narrow beam of radio waves. The beam rotates as the star spins. Using a radio telescope, an astronomer can detect a pulse of radio waves each time the pulsar rotates and the beam sweeps past the earth.

Quasars were first identified in 1963 by a group of astronomers at the Palomar Observatory, near San Diego. Since then, over a thousand have been discovered. Astronomers are not yet sure how quasars generate their vast quantities of radiation. Some believe quasars are powered by a giant blackhole that produces energy by swallowing clouds of gas from the surrounding galaxy.

Astronomers determine how distant a quasar is by measuring its red shift. Red shift indicates an object is moving away from the earth. According to Hubble’s law, the more distant the object is, the larger is its red shift. All quasars have large red shifts. Many astronomers believe quasars are the most distant objects yet detected in the universe. Some are estimated to be as far as 12 billion to 16 billion light-years from the earth.

$\boxed{\text{Assignment: 21.1, 21.2}}$

21.1 (Chinese version) $100 W/m^2 \longrightarrow 1340 W/m^2$

∇ (del) operator

In Cartesian coordinator system

$$\nabla = \boldsymbol{e}_x \frac{\partial}{\partial x} + \boldsymbol{e}_y \frac{\partial}{\partial y} + \boldsymbol{e}_z \frac{\partial}{\partial z}.$$

Descartes(笛卡尔) 1596-1690 French philosopher and mathematician

I think, therefore I exist.

I think, therefore iMac.

I am online, therefore I exist.

$$\operatorname{grad}\varphi = \nabla\varphi = \boldsymbol{e}_x \frac{\partial\varphi}{\partial x} + \boldsymbol{e}_y \frac{\partial\varphi}{\partial y} + \boldsymbol{e}_z \frac{\partial\varphi}{\partial z}.$$

$$\operatorname{div}\boldsymbol{A} = \nabla\cdot\boldsymbol{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}.$$

$$\operatorname{rot}\boldsymbol{A} = \nabla\times\boldsymbol{A} = \begin{vmatrix} \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\boldsymbol{e}_x + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\boldsymbol{e}_y + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\boldsymbol{e}_z.$$

Laplacian: $$\nabla^2 = \nabla\cdot\nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

$$\nabla = \boldsymbol{e}_x \frac{\partial}{\partial x} + \boldsymbol{e}_y \frac{\partial}{\partial y} + \boldsymbol{e}_z \frac{\partial}{\partial z}$$ is a differential as well as vector operator.

$$\nabla \cdot (\varphi \ \boldsymbol{f}) = \varphi \nabla \cdot \boldsymbol{f} + \boldsymbol{f} \cdot \nabla \varphi \qquad \text{(H4)}$$

$$\begin{aligned}
\nabla \cdot (\varphi \ \boldsymbol{f}) &= \frac{\partial \varphi f_x}{\partial x} + \frac{\partial \varphi f_y}{\partial y} + \frac{\partial \varphi f_z}{\partial z} \\
&= \frac{\partial \varphi}{\partial x} f_x + \frac{\partial \varphi}{\partial y} f_y + \frac{\partial \varphi}{\partial z} f_z + \varphi \left(\frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z} \right) \\
&= \nabla \varphi \cdot \boldsymbol{f} + \varphi \nabla \cdot \boldsymbol{f}
\end{aligned}$$

$$\nabla\cdot(\varphi\ \boldsymbol{f})=\nabla_{f}\cdot(\varphi\ \boldsymbol{f})+\nabla_{\varphi}\cdot(\varphi\ \boldsymbol{f})$$
$$=\varphi\nabla\cdot\boldsymbol{f}+\nabla\varphi\cdot\boldsymbol{f}$$

$$\nabla\cdot(\boldsymbol{f}\times\boldsymbol{g})=\boldsymbol{g}\cdot\nabla\times\boldsymbol{f}-\boldsymbol{f}\cdot\nabla\times\boldsymbol{g} \qquad \text{(H5)}$$

$$\nabla\cdot(\boldsymbol{f}\times\boldsymbol{g})=\nabla_{f}\cdot(\boldsymbol{f}\times\boldsymbol{g})+\nabla_{g}\cdot(\boldsymbol{f}\times\boldsymbol{g})$$
$$=\boldsymbol{g}\cdot(\nabla_{f}\times\boldsymbol{f})-\boldsymbol{f}\cdot(\nabla_{g}\times\boldsymbol{g})$$
$$=\boldsymbol{g}\cdot\nabla\times\boldsymbol{f}-\boldsymbol{f}\cdot\nabla\times\boldsymbol{g}$$

$$\nabla\times(\nabla\times f)=\nabla(\nabla\cdot f)-\nabla^2 f. \qquad \text{(H8)}$$

$$\nabla\times(\nabla\times f)=\nabla(\nabla\cdot f)-\nabla\cdot\nabla f$$

$$=\nabla(\nabla\cdot f)-\nabla^2 f.$$

$$\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b})=(\boldsymbol{c}\cdot\boldsymbol{b})\boldsymbol{a}-(\boldsymbol{c}\cdot\boldsymbol{a})\boldsymbol{b}=\boldsymbol{a}(\boldsymbol{c}\cdot\boldsymbol{b})-\boldsymbol{b}(\boldsymbol{c}\cdot\boldsymbol{a}).$$

Set $\boldsymbol{d}=\boldsymbol{a}\times\boldsymbol{b},\ \boldsymbol{f}=\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b}).$

$$f_x=c_y d_z-c_z d_y$$

$$=c_y(a_x b_y-a_y b_x)-c_z(a_z b_x-a_x b_z)$$

$$=a_x(c_y b_y+c_z b_z)-b_x(c_y a_y+c_z a_z)$$

$$=a_x(c_x b_x+c_y b_y+c_z b_z)-b_x(c_x a_x+c_y a_y+c_z a_z)$$

$$=a_x(\boldsymbol{c}\cdot\boldsymbol{b})-b_x(\boldsymbol{c}\cdot\boldsymbol{a})$$

The gradient of a scalar field is a rotation-free vector field.

$$\nabla \times \nabla \varphi = 0.$$

Proof: Set $\boldsymbol{f} = \nabla \varphi.$

$$(\nabla \times \nabla \varphi)_x = (\nabla \times \boldsymbol{f})_x = \frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z}$$

$$= \frac{\partial}{\partial y}\left(\frac{\partial \varphi}{\partial z}\right) - \frac{\partial}{\partial z}\left(\frac{\partial \varphi}{\partial x}\right) = 0.$$

Similarly, we can prove that the other components are zero too. Therefore $\nabla \times \nabla \varphi = 0.$

The rotation of a vector field is a vector field without source.

$$\nabla \cdot \nabla \times \boldsymbol{f} = 0.$$

Proof:

$$\nabla \cdot \nabla \times \boldsymbol{f}$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial f_x}{\partial z} - \frac{\partial f_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y}\right)$$

$$= 0.$$

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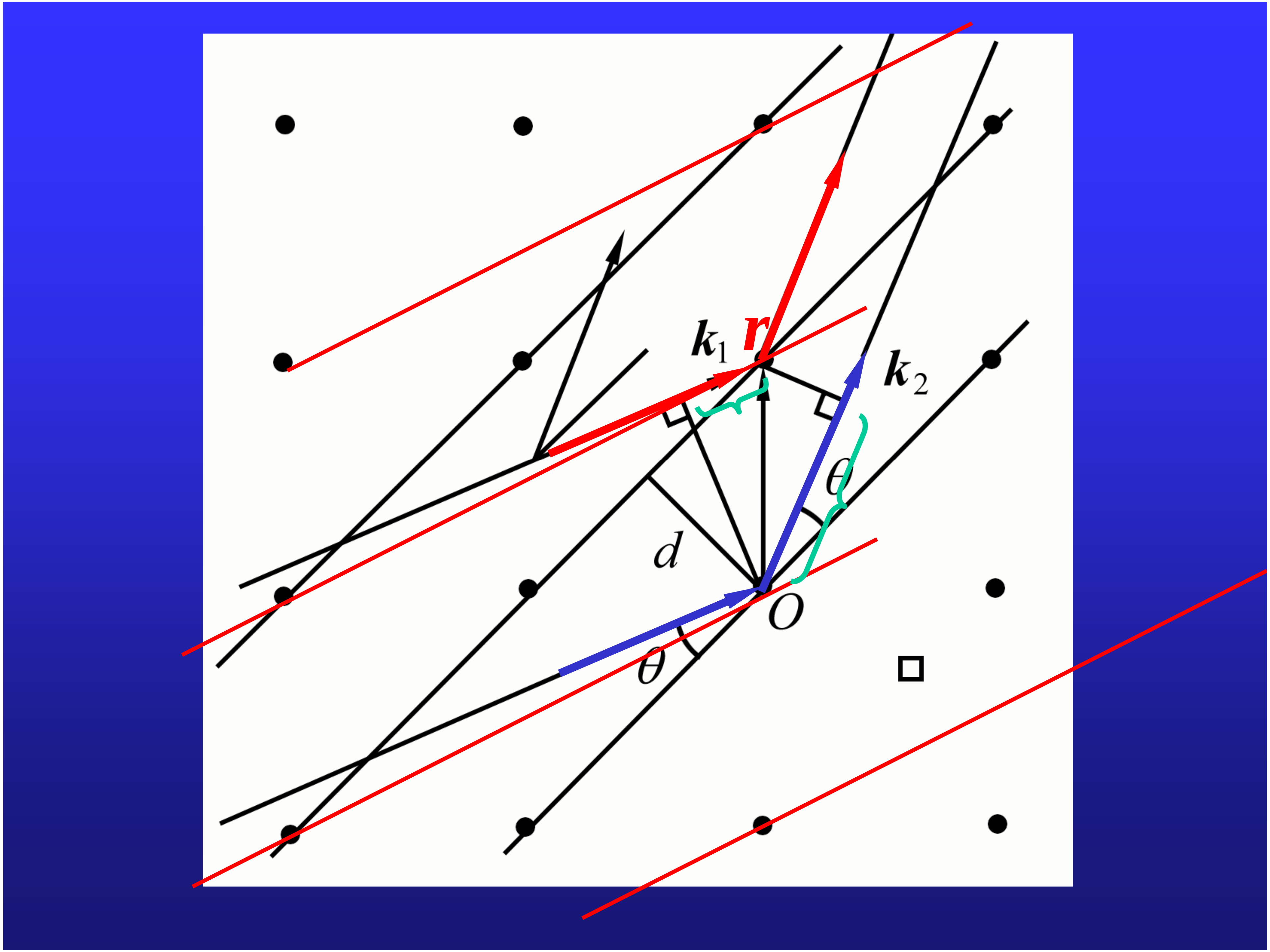
21.4 Crystal diffraction

X rays: $\lambda \sim$ 1nm or 1Å

In a crystal, the atom spacing ~ 1Å

X rays scattered by atoms or atom groups，scattering waves interfere with each other



k1
r
k2
θ
d
θ
O

The path difference of scattering waves from adjacent planes is

$$\hat{\boldsymbol{k}}_2 \cdot \boldsymbol{r} - \hat{\boldsymbol{k}}_1 \cdot \boldsymbol{r}$$

Constructive interference requires

$$\hat{\boldsymbol{k}}_2 \cdot \boldsymbol{r} - \hat{\boldsymbol{k}}_1 \cdot \boldsymbol{r} = n\lambda$$

$$\boldsymbol{k}_2 \cdot \boldsymbol{r} - \boldsymbol{k}_1 \cdot \boldsymbol{r} = 2n\pi$$

Since

$$\left(\boldsymbol{k}_2 \cdot \boldsymbol{r} - \boldsymbol{k}_1 \cdot \boldsymbol{r}\right) = \Delta \boldsymbol{k} \cdot \boldsymbol{r} = \Delta k \, d$$

$$\Delta k = 2k \sin\theta = \frac{4\pi}{\lambda} \sin\theta$$

Condition for constructive interference is

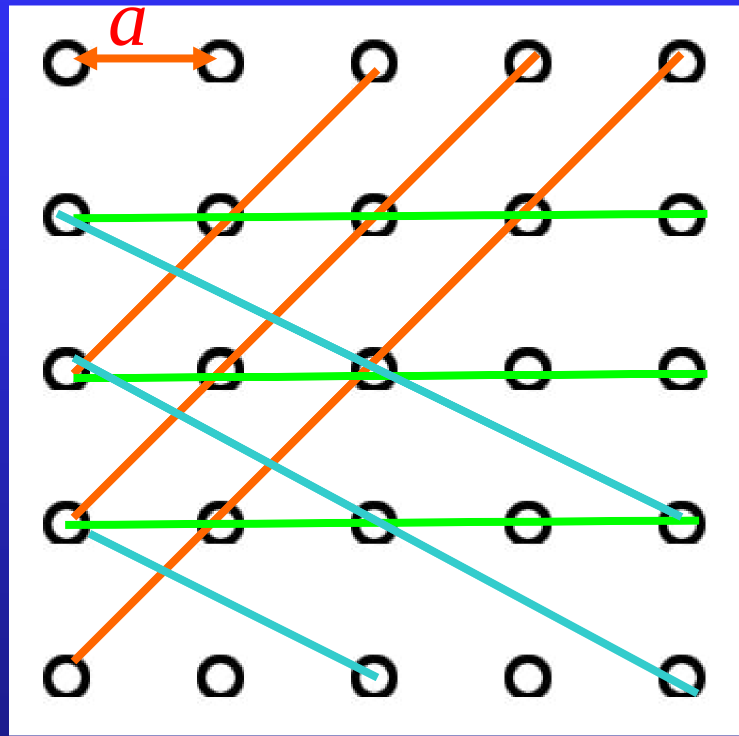
$$2d\sin\theta = n\lambda$$

between wave vector and crystal plane

a-lattice constant

families of parallel planes

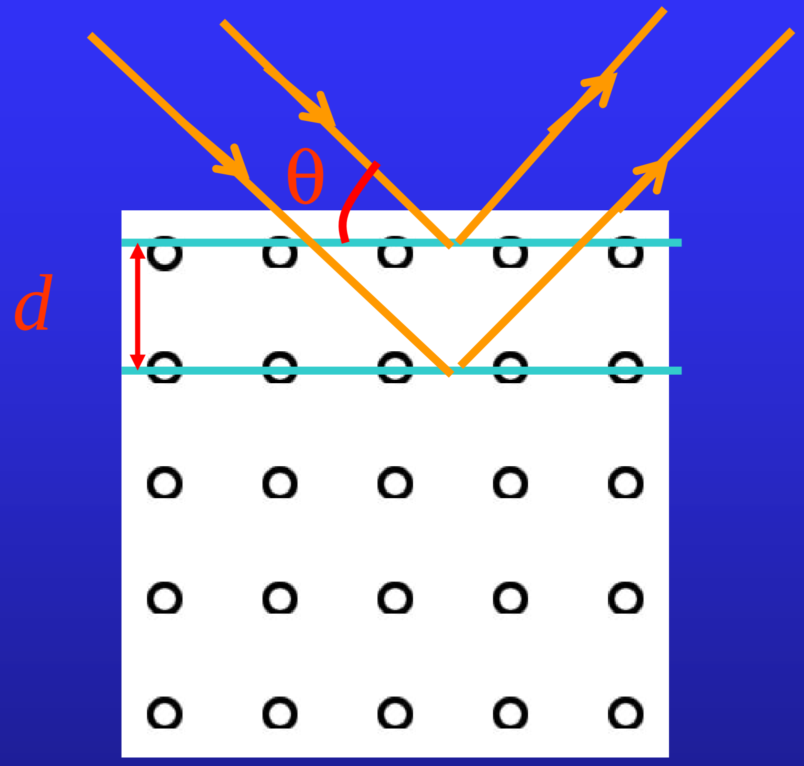
——crystal planes



A portion of beam of X-ray is reflected from the front plane, and a portion of beam is reflected from the second plane, and so forth.

The path difference of two reflected waves is

$$2d\sin\theta.$$

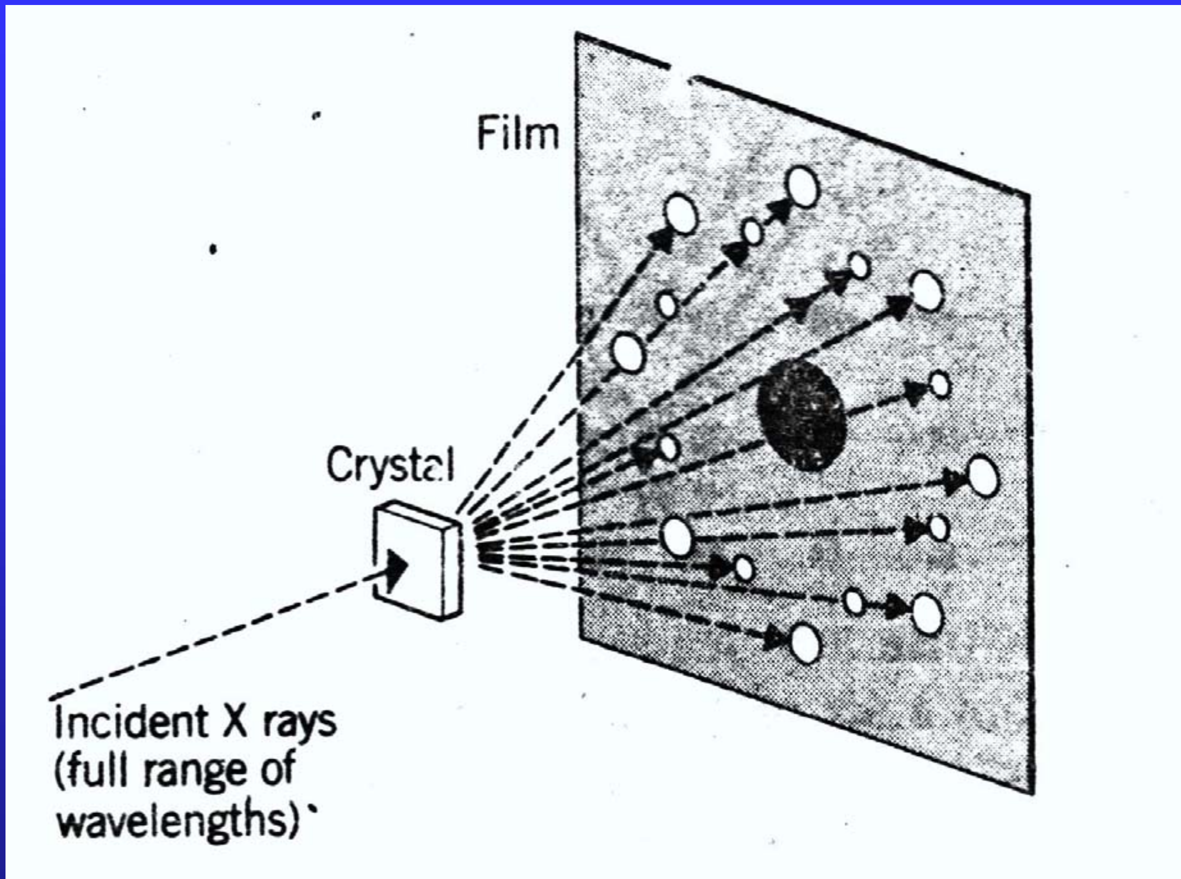


The condition for constructive interference is

$$2d\sin\theta = n\lambda, \qquad n = 1,2,3,\cdots$$

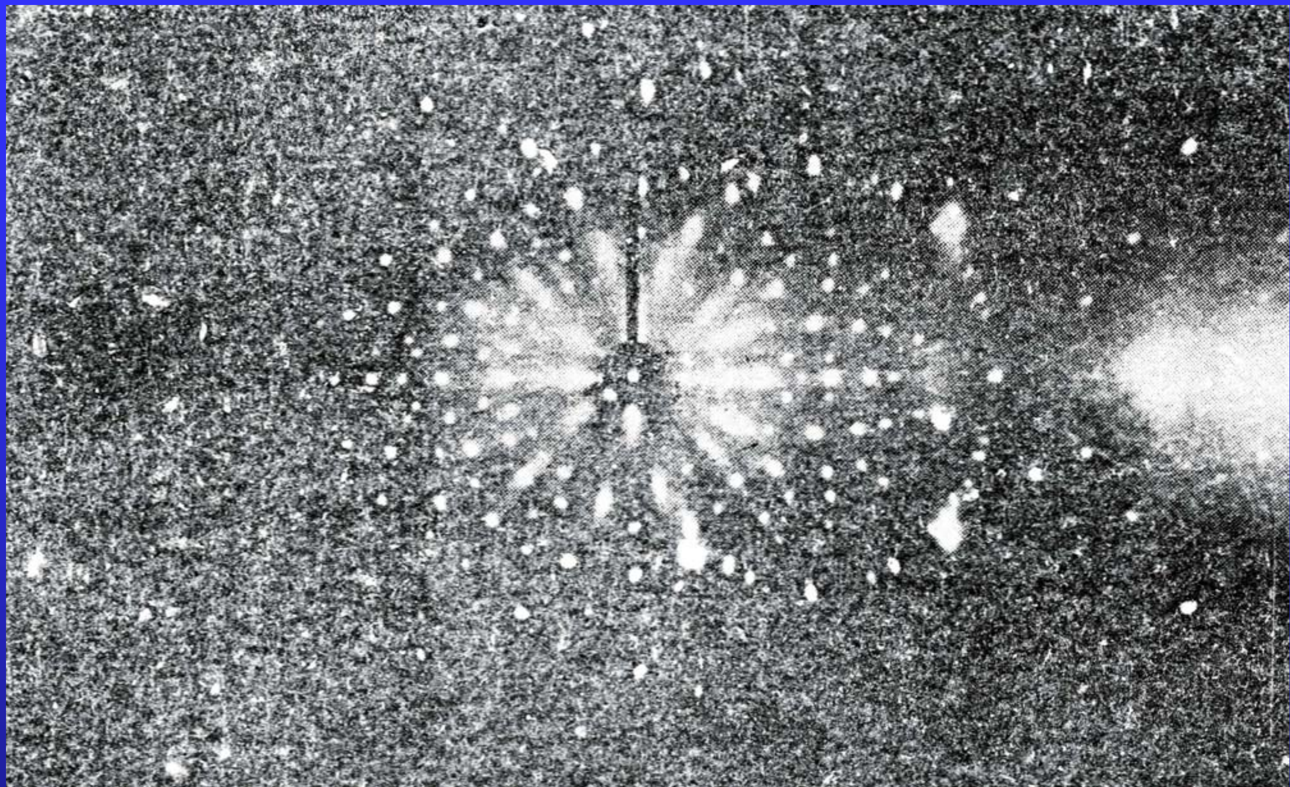
Bragg's condition

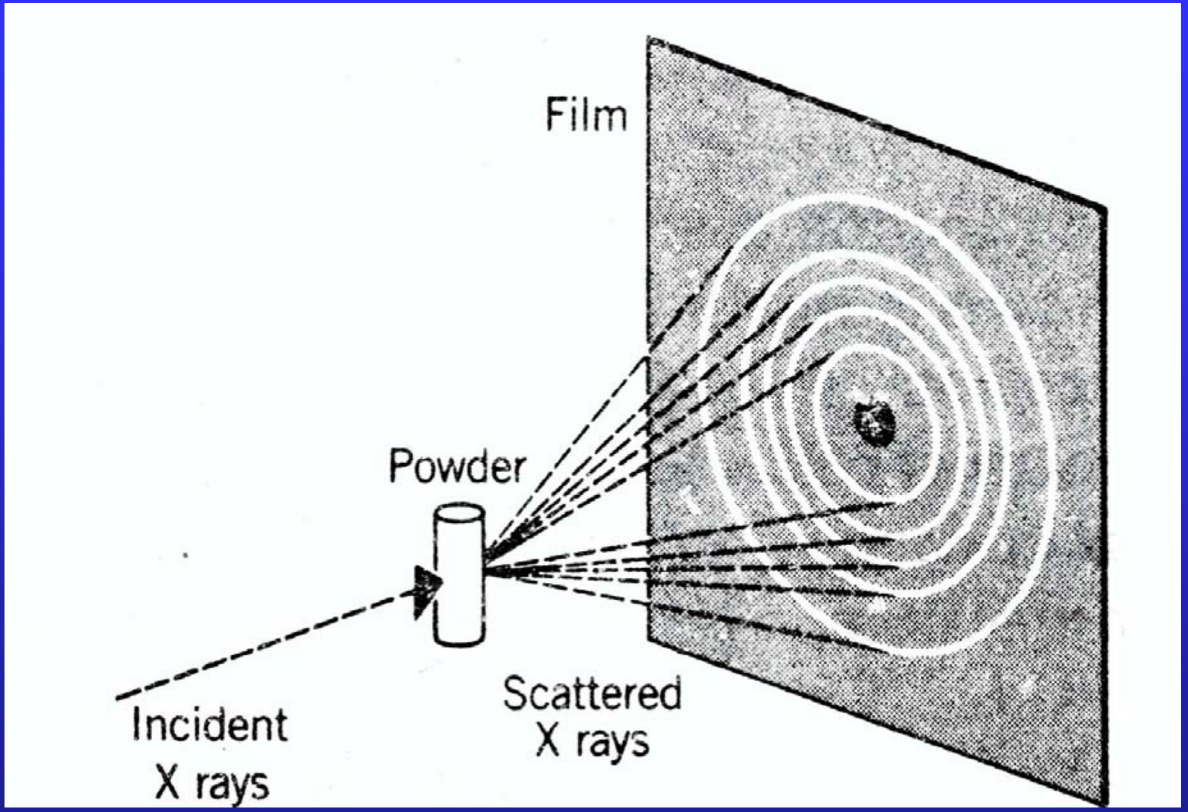
X-rays diffraction is powerful tool of structure analysis.



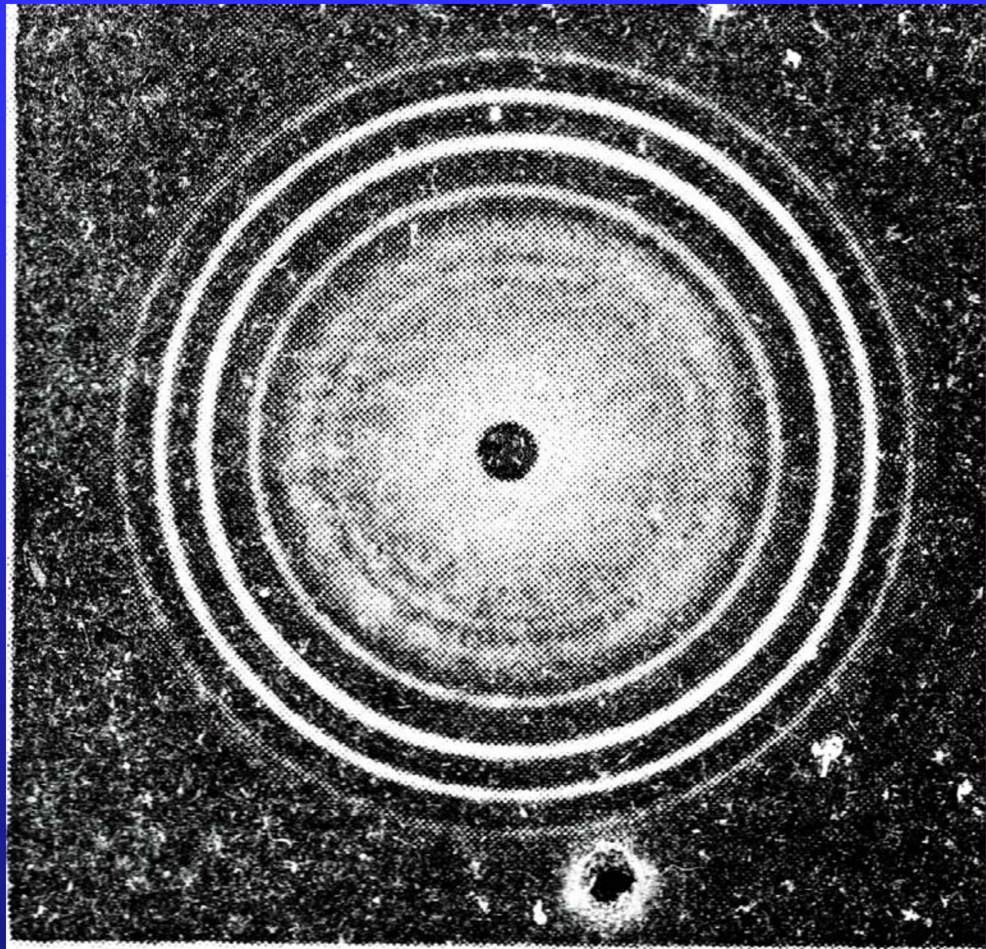
diffraction
pattern

The Laue
pattern





The Debye pattern



X-ray diffraction and the discovery of the DNA structure

Nitrogen base
含氮碱基

A G G C T T A

— S — P — S — P — S — P — S — P — S — P — S — P — S —

sugar(deoxyribose)
脱氧核糖

Phosphate
磷酸

A(adenine) 腺嘌呤， G(guanine) 鸟嘌呤
T(thymine) 胸腺嘧啶 C(cytosine) 胞嘧啶

adeno: latin for “gland” guano:dung of sea birds or bats
thymus：胸腺 cyt, cyto pertaining to cells

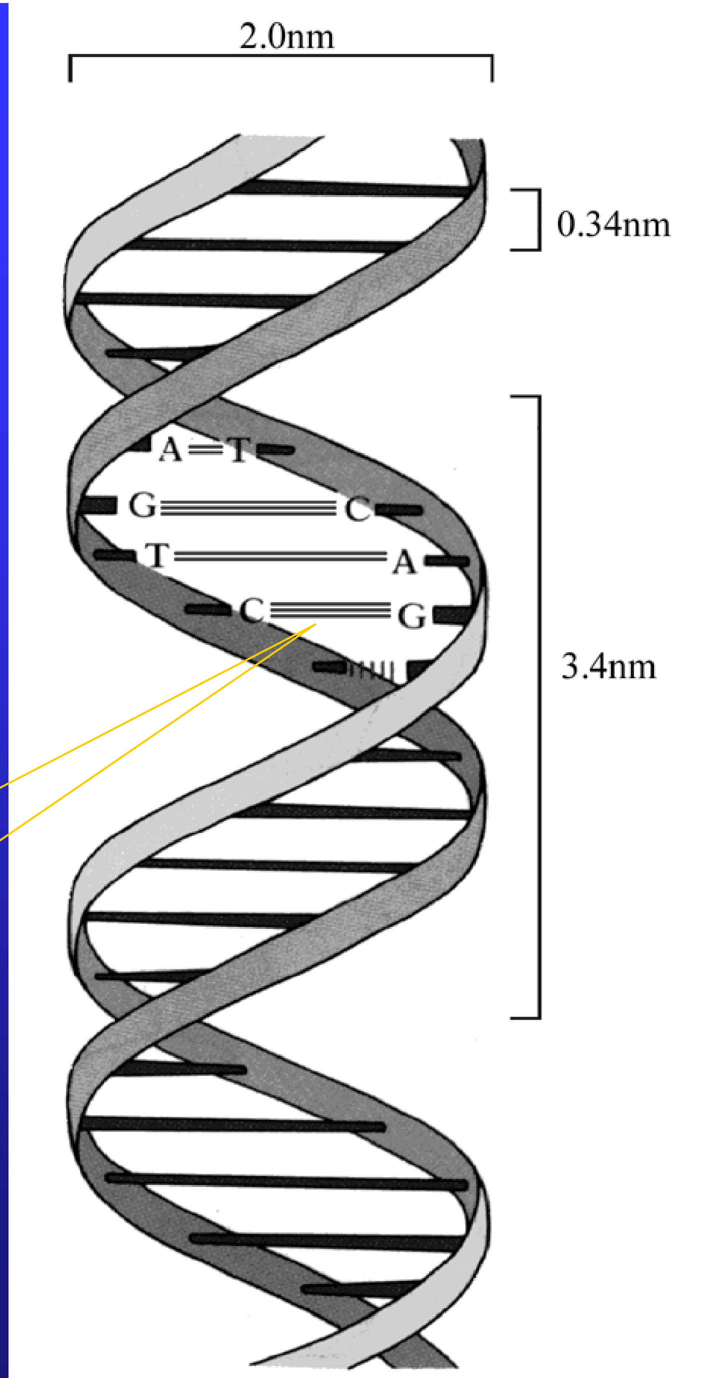
double helix structure

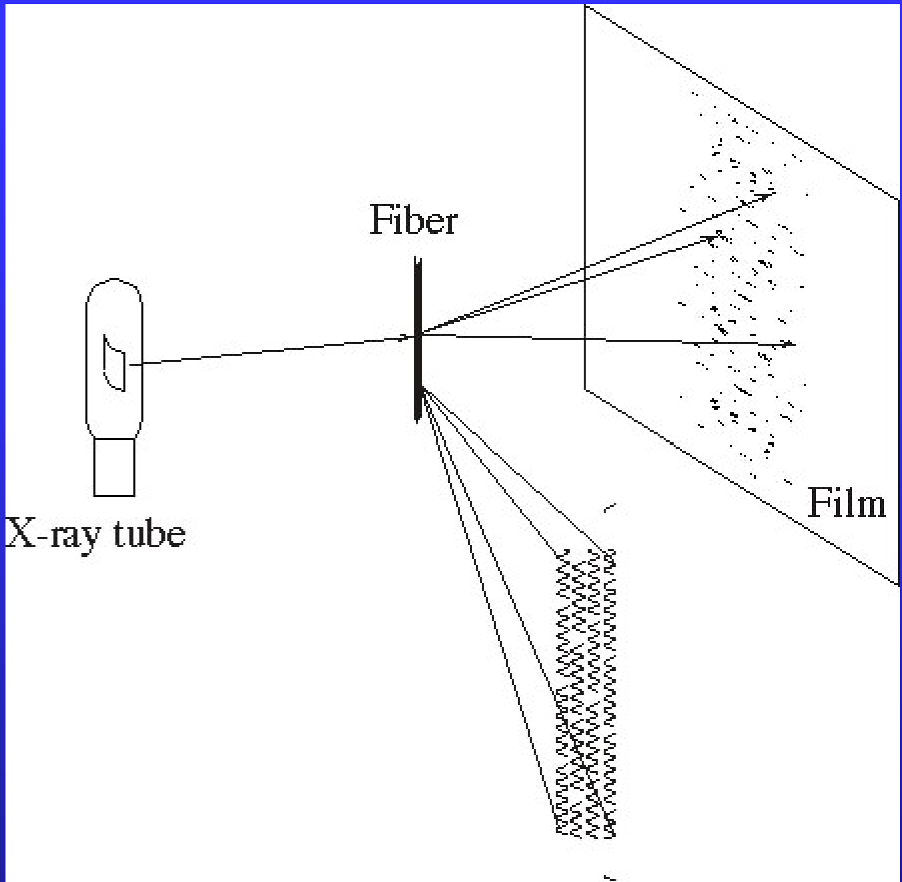
complementary rule:

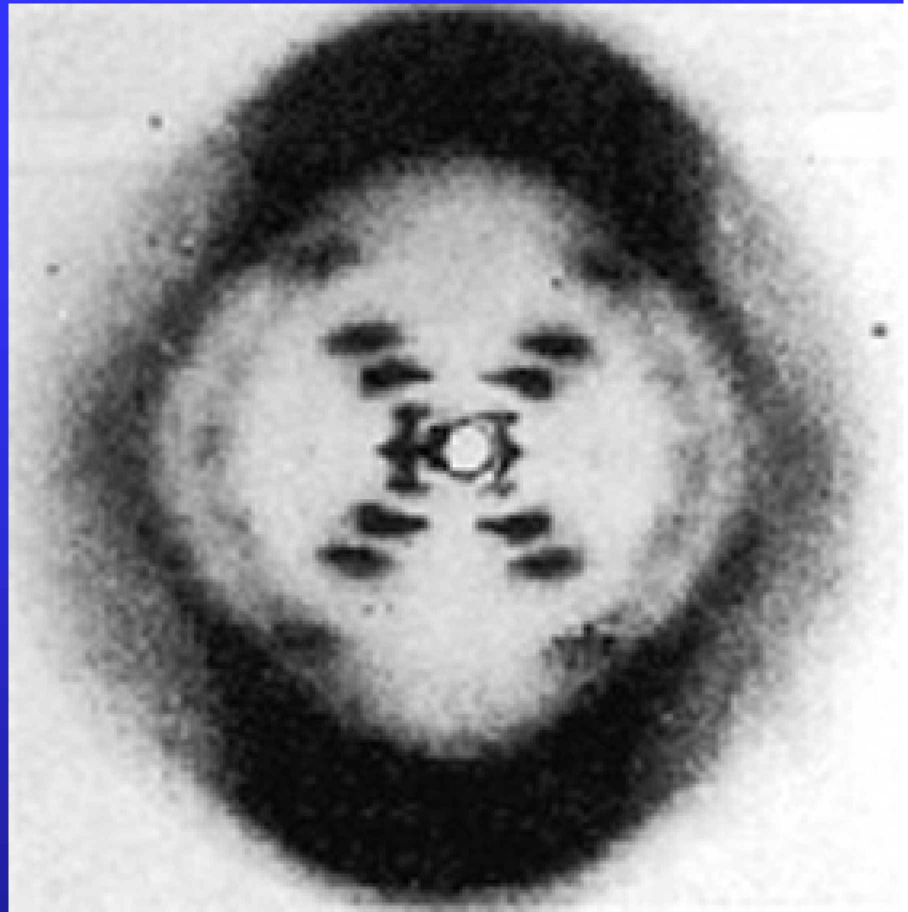
$$A = T,\ G \equiv C,\ T = A,\ C \equiv G$$

the template theory

Hydrogen bond

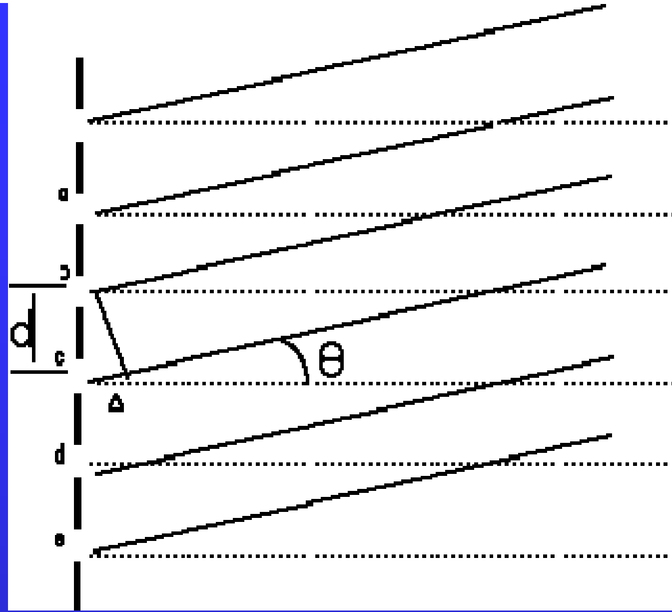


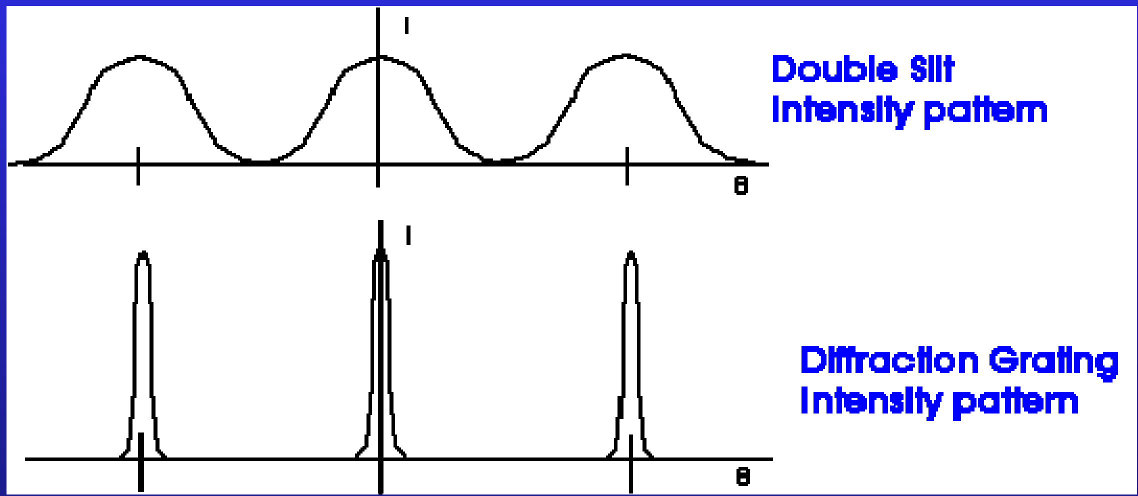




- X-shaped cross
- The fourth layer line of diffraction is missing

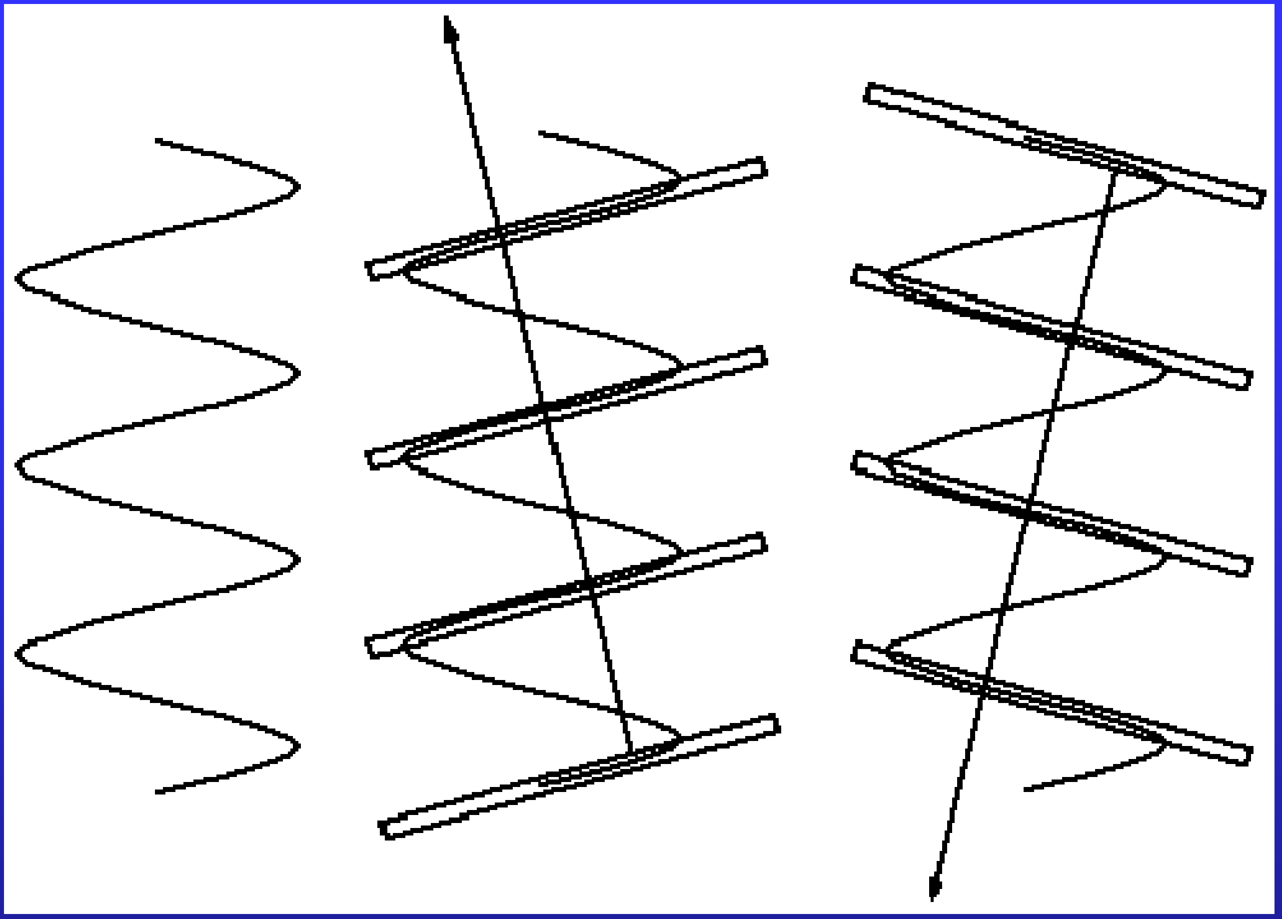
diffraction grating



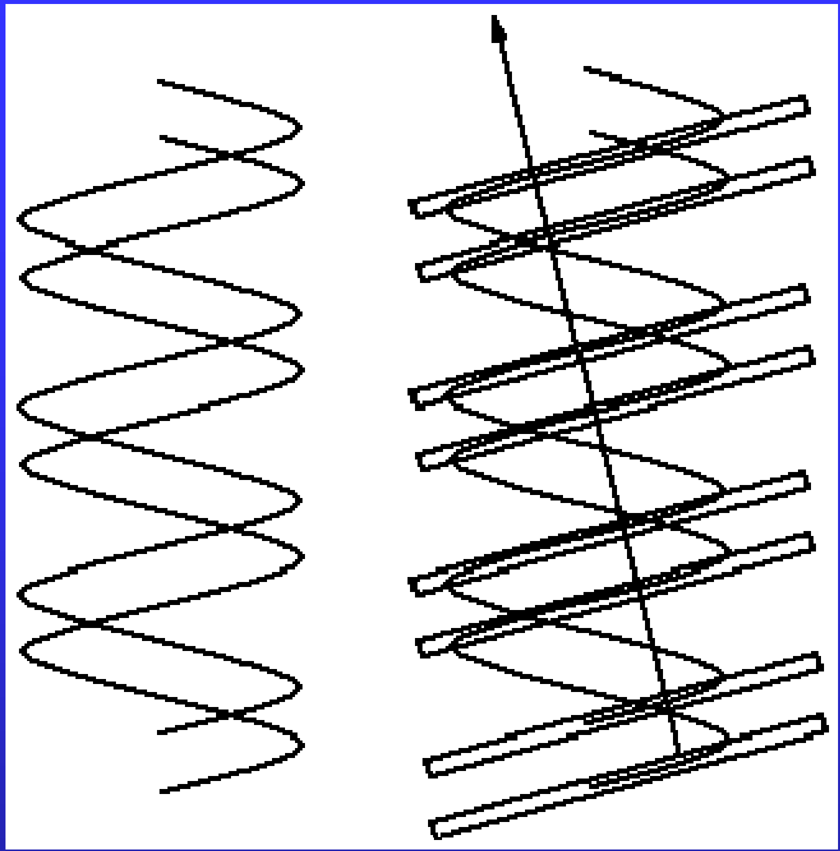


constructive interference for $d \sin\theta = m\lambda,$

$$\mathrm{m} = 0, \pm 1, \pm 2, \pm 3 \cdots$$



•X-shaped Cross



double slit interference

$$\sin\theta = m'\frac{\lambda}{d'}$$ constructive

$$\sin\theta = \left(m' + \frac{1}{2}\right)\frac{\lambda}{d'}$$ destructive

$$d' = \frac{3}{8} d, \qquad m' = 1 \qquad m' = -2 \qquad \sin\theta = \pm \frac{3}{2} \frac{\lambda}{d'} = \pm 4 \ \frac{d}{\lambda}$$

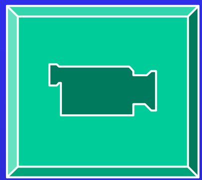
destructive

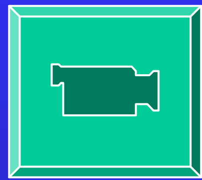
Just the position of the fourth layer line

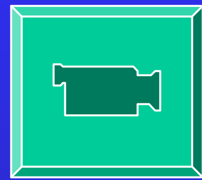
$$\sin\theta = m \frac{\lambda}{d}, \qquad m = \pm 4 \qquad \sin\theta = \pm 4 \ \frac{d}{\lambda}$$

$$m = 0, \pm 1, \pm 2, \pm 3 \cdots$$

Amand Lucas University of Namur (Belgium)



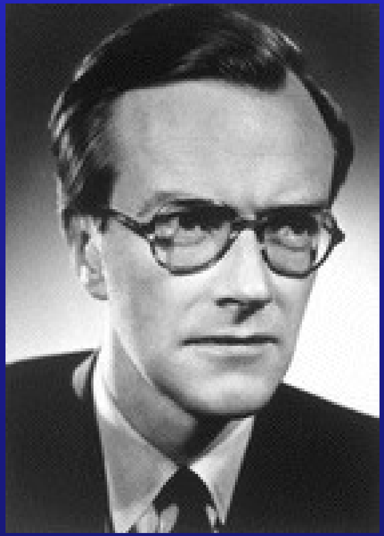




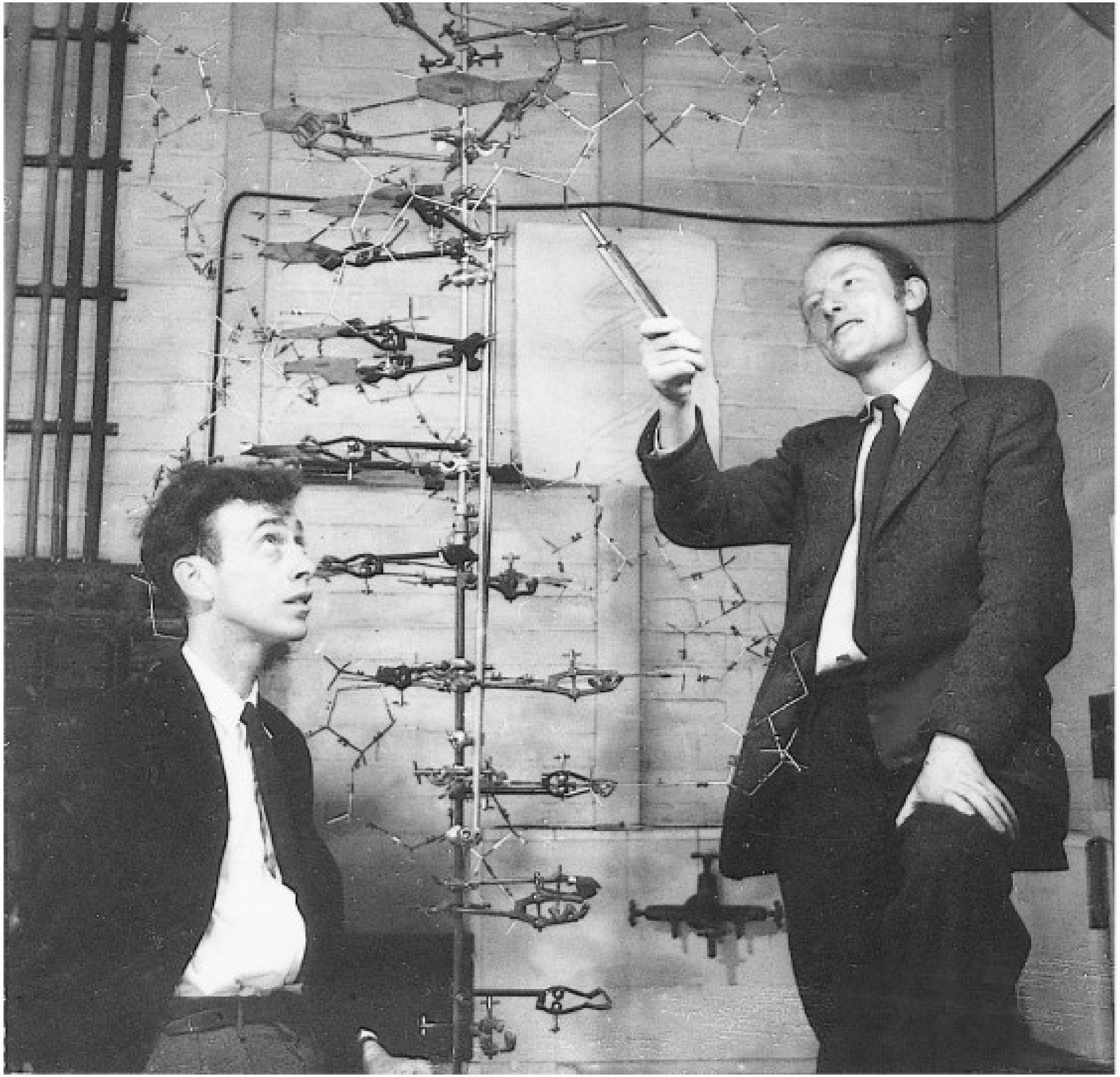
Rosalind Franklin

James D. Watson

Francis Crick



Maurice Wilkins



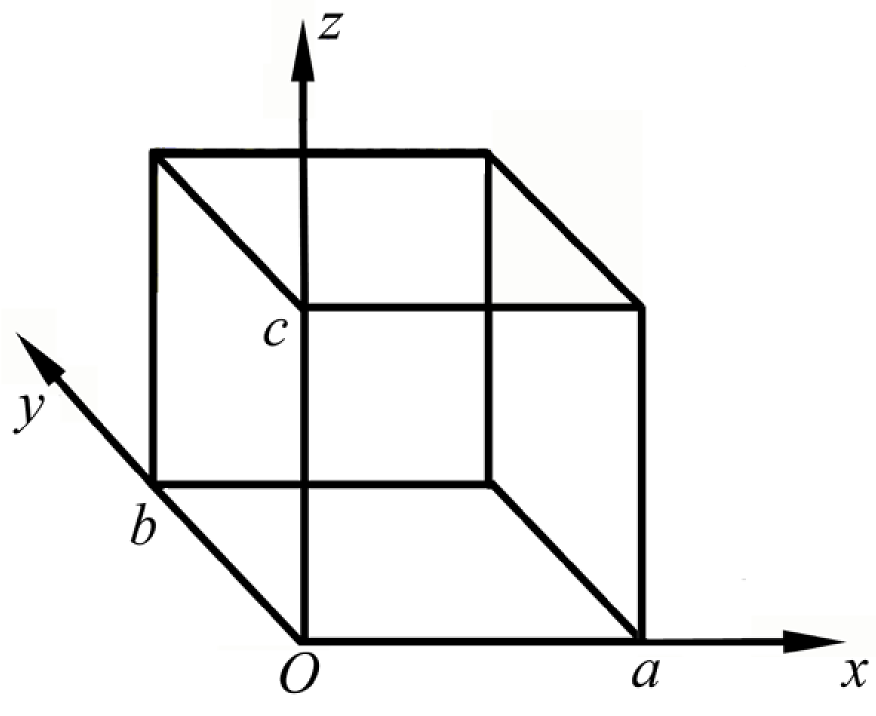
21.5 Standing wave and DOS

A monochrome plane wave can be write as

$$\boldsymbol{E} = \boldsymbol{E}(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\omega t}$$

Wave equation for one component E_x is

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + \frac{\omega^2}{c^2}\right)E_x = 0$$



Relative to $y = 0, b$; $z = 0, c$ surfaces, E_x is the tangent component

$$E_x = E_{x0}\; f_1(x) \boxed{\sin\left(\frac{m\pi}{b} y\right)} \boxed{\sin\left(\frac{n\pi}{c} z\right)}$$

$$f_1(x) = C_1 \sin k_1 x + C_1' \cos k_1 x$$

$$\nabla \cdot \boldsymbol{E} = 0 \qquad E_x = E_{x0} \boxed{\cos\left(\frac{\ell\pi}{a} x\right)} \sin\left(\frac{m\pi}{b} y\right) \sin\left(\frac{n\pi}{c} z\right)$$

E_y and E_z has same structure.

Substituting into wave equation yields

$$k^2 \equiv \frac{\ell^2\pi^2}{a^2} + \frac{m^2\pi^2}{b^2} + \frac{n^2\pi^2}{c^2} = \frac{\omega^2}{c^2}$$

Set $a = b = c = L$

$$k^2 = \frac{\pi^2}{L^2}(\ell^2 + m^2 + n^2)$$

A certain set of (ℓ, m, n) ~ a special wave mode,
or a "state"

$k \longleftrightarrow$ several states

$$k = \frac{\pi}{L}\sqrt{6} \longleftrightarrow (2,1,1),\ (1,2,1) \text{ and } (1,1,2)$$

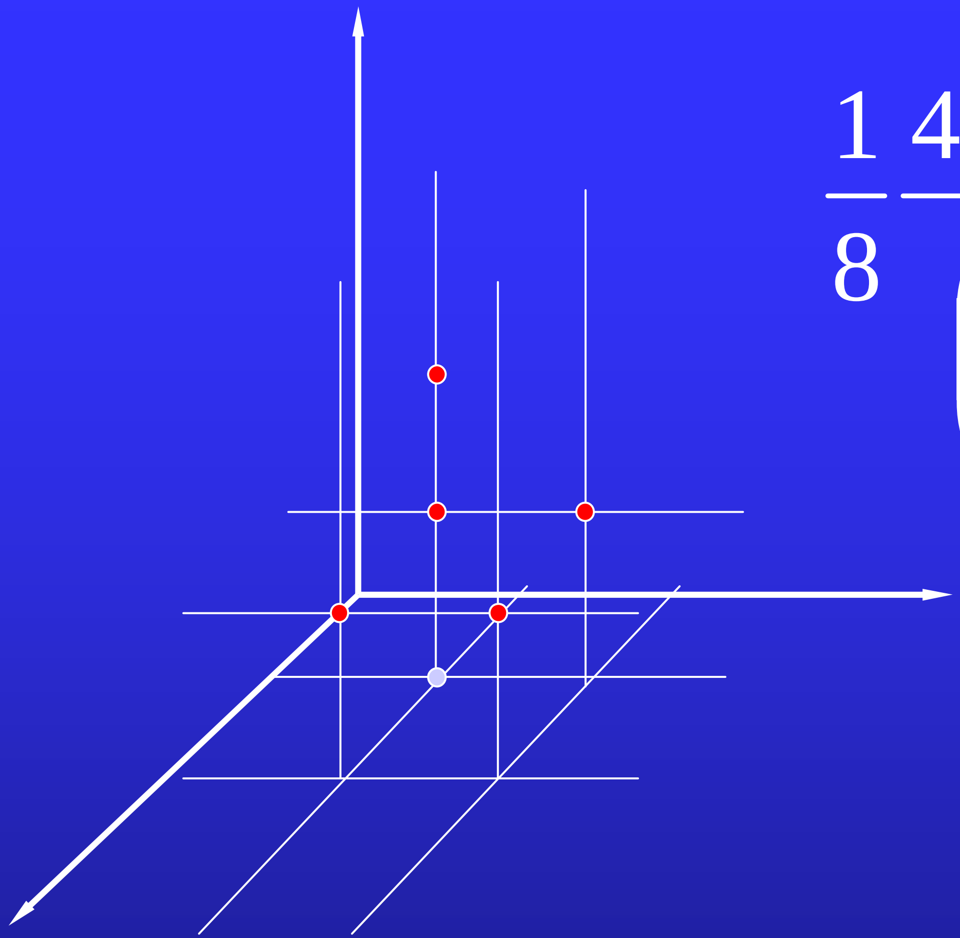
How many states are there in the interval k to k + dk**?**

Image a k-space.

- The unit of length is π/L.
- A set of (ℓ, m, n) represents a point in the first

quadrant of the space

- On average a point occupies a volume $(\pi/L)^3$



$$\frac{1}{8}\frac{4\pi k^2 \mathrm{d}k}{\left(\dfrac{\pi}{L}\right)^3} = V\frac{1}{2\pi^2}k^2\mathrm{d}k$$

$$\frac{V}{2\pi^2 c^3}\omega^2\mathrm{d}\omega = \frac{4\pi V}{c^3}\nu^2\mathrm{d}\nu$$

density of states (DOS) D

$$D\,\mathrm{d}\nu = \frac{4\pi}{c^3}\nu^2\mathrm{d}\nu$$

Electromagnetic wave has two polarization directions

$$D\,\mathrm{d}\nu = \frac{8\pi}{c^3}\nu^2\mathrm{d}\nu$$